

Lightcone quantisation of the bosonic string

Abstract

We introduce the classical string by solving the Polyakov action in the lightcone gauge which is useful in encompassing the non-linear constraints that arise from the action. The drawback of the lightcone gauge is that it is not manifestly Lorentz invariant. We show that it is only in $D = 26$ that Lorentz invariance is restored. We study the spectrum of the strings up to level two and introduce DDF states to help us build the physical spectrum of the strings, that is free of ghosts. We conclude the report by proving that the spectrum is indeed free of ghosts in $D = 26$ spacetime dimensions.

1 The Classical String

We start our exposition by recalling setting up the action and solving for the equations of motion and symmetries of the classical string. These will be needed explicitly when we make arguments towards fixing the gauge of the classical string before quantisation. We will study the string in a background spacetime $\mathcal{M} = \mathbf{R}^{D-1,1}$, also called target space, which is a D -dimensional Minkowski spacetime. More general curved backgrounds are possible, but we restrict attention to the simplest case. The background metric is denoted by $G_{\mu\nu} \equiv \eta_{\mu\nu}$.

Embedded in the target space is a string, which traverses a *worldsheet*, Σ ; a two dimensional surface parametrised by a timelike coordinate τ and a spacelike coordinate σ , which we compactly denote by $\sigma^a = (\tau, \sigma)$, $a = 0, 1$. We use spacetime coordinates $X^\mu(\tau, \sigma)$, $\mu = 0, \dots, D-1$. Given the background metric $\eta_{\mu\nu}$, the induced metric on the worldsheet is a two-dimensional dynamical quantity given by,

$$h_{ab} = \eta_{\mu\nu} \partial_a X^\mu \partial_b X^\nu$$

Motivated by the action of a classical relativistic particle, we may write the action of the string being proportional to the area traced out by the worldsheet,

$$S_{NG} = -T \int_{\Sigma} \sqrt{-h} d^2\sigma$$

where $h = \det(h_{ab})$ and the prefactor is a measure of the tension of the string given by, $T = \frac{1}{2\pi\alpha'}$ with α' called the Regge slope. From now on, we set $\alpha' = \frac{1}{2}$ which will simplify some equations in the next sections. S_{NG} is the so called *Nambu-Goto* action. We can write it in coordinate form by explicitly calculating the induced metric on Σ ,

$$h_{00} = (\partial_\tau X)^2, \quad h_{11} = (\partial_\sigma X)^2, \quad h_{01} = h_{10} = (\partial_\tau X) \cdot (\partial_\sigma X)$$

so that,

$$S_{NG} = -T \int_{\Sigma} \sqrt{-(\partial_\tau X)^2 (\partial_\sigma X)^2 + (\partial_\tau X)^2 (\partial_\sigma X)^2} d^2\sigma \quad (1)$$

The drawback of the Nambu-Goto action is that it has a non-polynomial form, making it very hard to calculate with. The route to avoid that is to instead use the *Polyakov* action, by introducing a dynamical field, γ_{ab} coupled to the scalar fields X^μ on the worldsheet. Polyakov's action is,

$$S_P = -\frac{T}{2} \int d^2\sigma \sqrt{-\gamma} \gamma^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu \eta_{\mu\nu}. \quad (2)$$

The metric on the worldsheet is a new dynamical quantity which has its own equation of motion which can be obtained varying the action with respect to $\gamma^{\rho\sigma}$. Requiring that for the classical solutions $\delta S_P = 0$ with respect to metric variations is equivalent to the vanishing of the stress-energy tensor. Explicitly,

$$T_{\alpha\beta} = -\frac{2}{T} \frac{1}{\sqrt{-\gamma}} \frac{\delta S_P}{\delta \gamma^{\alpha\beta}} = 0$$

Using the identity $\delta\sqrt{-\gamma} = \frac{1}{2}\sqrt{-\gamma}\gamma^{ab}\delta\gamma_{ab} = -\frac{1}{2}\sqrt{-\gamma}\gamma_{ab}\delta\gamma^{ab}$, we have

$$\frac{\delta S_P}{\delta \gamma^{\rho\sigma}} = -\frac{T}{2} \int d^2\sigma \sqrt{-\gamma} \left\{ h_{\rho\sigma} - \frac{1}{2} \gamma_{\rho\sigma} \gamma^{\alpha\beta} h_{\alpha\beta} \right\}$$

with $h_{\alpha\beta}$ the induced metric on Σ . Therefore the equation of motion of γ is,

$$h_{\rho\sigma} = \frac{1}{2} \gamma_{\rho\sigma} \gamma^{\alpha\beta} h_{\alpha\beta} \quad (3)$$

which is solved by,

$$\gamma_{\alpha\beta} = f(\sigma, \tau) h_{\alpha\beta}$$

for any function $f(\sigma, \tau)$ and $h_{\alpha\beta}$. Substituting the solution back to the Polyakov action we obtain the Nambu-Goto action, showing therefore that the two are equivalent when the EOM are satisfied.

For the classical solutions X^μ , we vary the Polyakov action with respect to X^μ and obtain the equation of motion,

$$\partial_\alpha (\sqrt{-\gamma} \gamma^{\alpha\beta} \partial_\beta X^\mu) = 0 \quad (4)$$

We will shortly simplify the equation of motion for X^μ by fixing the gauge, but before doing that we need to think about the symmetries of the action. It turns out that we have three different classes of symmetry, and that they are enough to almost completely solve for X^μ .

1. S_P has global Poincare invariance, which means that the action is invariant under translations ($X^\mu \rightarrow X^\mu + a^\mu$ for some constant vector a^μ) and Lorentz transformations ($X^\mu \rightarrow \Lambda^\mu_\nu X^\nu$).
2. S_P is invariant under local diffeomorphisms. To understand this we first note that it should be expected that this is the case: the action is ignorant on how we choose to parametrise it (coordinates σ^a). Under $\sigma^a \rightarrow \tilde{\sigma}^a$, the metric and fields X^μ transform as,

$$\begin{aligned} X^\mu(\sigma) &\rightarrow \tilde{X}^\mu(\tilde{\sigma}) = X^\mu(\sigma) \\ \gamma_{\alpha\beta}(\sigma) &\rightarrow \tilde{\gamma}_{\alpha\beta}(\tilde{\sigma}) = \frac{\partial \sigma^\kappa}{\partial \tilde{\sigma}^\alpha} \frac{\partial \sigma^\lambda}{\partial \tilde{\sigma}^\beta} \gamma_{\kappa\lambda}(\sigma) \end{aligned}$$

3. Finally the action is also invariant under a local Weyl rescaling of the metric,

$$\gamma_{\alpha\beta} \rightarrow e^{2g(\sigma)} \gamma_{\alpha\beta}$$

where $g(\sigma)$ is a function of the coordinates (σ, τ) .

We claimed above that the symmetries will be enough to fix the gauge and solve for the classical solutions. To see why this is true, recall that the metric is a symmetric bilinear form in two-dimensions, and therefore has 3 degrees of freedom. Invariance under local diffeomorphisms allows us to fix two of them (eg, choosing two functions for $\sigma^a \rightarrow \tilde{\sigma}^a$) and the third one is fixed

by choosing a single function $g(\sigma)$ and performing a Weyl rescaling. Using these symmetries we can simply set the metric to the flat one, $\gamma_{\alpha\beta} = \eta_{\alpha\beta}$, simplifying the EOM significantly.

The EOM for the field X^μ is,

$$X^\mu \partial_\tau^2 X^\mu - \partial_\sigma^2 X^\mu = 0 \quad (5)$$

and the EOM for the dynamical metric arising from the vanishing of the various components of the stress-energy tensor are,

$$\begin{aligned} T_{00} = T_{11} &= \frac{1}{2} ((\partial_\tau X)^2 + (\partial_\sigma X)^2) = 0 \\ T_{01} = T_{10} &= (\partial_\tau X) \cdot (\partial_\sigma X) = 0 \end{aligned} \quad (6)$$

The EOM for X^μ is a wave equation in (σ, τ) , for which the solutions will be sums of right and left moving modes. The latter two equations are called the *Virasoro constraints* and are more difficult to solve, being non-linear. This is where the lightcone gauge will prove to be very powerful. The most general solution of X^μ is given by,

$$X^\mu(\sigma, \tau) = x^\mu + v^\mu \tau + X_R^\mu(\tau - \sigma) + X_L^\mu(\tau + \sigma) \quad (7)$$

where x^μ, v^μ are constant vectors that can be identified as the centre of mass position and velocity of the string, and $X_{R,L}$ are right and left moving waves of a single variable as indicated explicitly.

Before moving to the exploration of the lightcone gauge, we write the solution of X^μ based on the boundary conditions. For an open string, $X_L^\mu = X_R^\mu$, we let $\sigma \in [0, \pi]$ and get two different possibilities at the boundary,

- Dirichlet boundary conditions, $\delta X^\mu(0, \tau) = \delta X^\mu(\pi, \tau) = 0$
- Neumann boundary conditions, $\partial_\sigma X^\mu(0, \tau) = \partial_\sigma X^\mu(\pi, \tau) = 0$

For a closed string, X_L^μ and X_R^μ are independent, we let $\sigma \in [0, 2\pi]$ and require that X^μ is a periodic function, $X^\mu(\sigma, \tau) = X^\mu(\sigma + 2\pi, \tau)$.

2 Fixing to lightcone gauge

In the old covariant quantisation, after solving the equation of motion for X^μ , we quantise the string by promoting the mode coefficients to operators and then imposing commutation relations among the creation and annihilation operators. The Virasoro constraints are then applied to the big Hilbert space to reduce it to that of physical states.

In the lightcone gauge we take a different route: we will make a smart choice for the gauge in order to exactly solve for the field X^μ still at the classical level and only then we promote the modes to operators. The issue that we will need to deal with is that the gauge choice is not manifestly Lorentz invariant and we will have to find the necessary and sufficient conditions for which it is. One of the conditions will be that the critical dimension of the string is $D = 26$.

The first step in quantising with lightcone gauge is to realise that there is a leftover gauge freedom that we can use to our advantage. We first introduce lightcone coordinates on the worldsheet by defining $\sigma^\pm = \tau \pm \sigma$. The metric in the new coordinates takes the simpler form, $ds^2 = -d\sigma^+ d\sigma^-$. Consider now the reparametrizations,

$$\sigma^+ \rightarrow \sigma'^+ = F(\sigma^+) \quad \sigma^- \rightarrow \sigma'^- = G(\sigma^-)$$

under which the metric takes the form,

$$ds'^2 = -d\sigma'^+ d\sigma^- = -\partial_+ F \cdot \partial_- G d\sigma^+ d\sigma^-$$

With X^μ as written above in Eq. 7, we can see that it is actually a function of lightcone coordinates. Therefore, if we choose one of the parameters, eg. τ' to be given by,

$$\tau' = F(\tau + \sigma) + G(\tau - \sigma),$$

for functions $F(\sigma^+), G(\sigma^-)$ that are combinations of X_R^μ and X_L^μ , our gauge freedom corresponds to reparametrising τ' so that it is equal to one of the X^μ . To see this, we first introduce lightcone coordinates on the target space also,

$$X^\pm = \frac{X^0 \pm X^{D-1}}{\sqrt{2}}$$

so that $X^\mu = (X^+, X^i, X^-)$ and the metric is given by $ds^2 = -2dX^+ dX^- + dX^i dX^i$ (summation convention assumed in $i = 1, \dots, D-2$). The lightcone gauge is then,

$$X^+(\sigma, \tau) = x^+ + p^+ \tau \tag{8}$$

for some functions x^+, p^+ . Since the lightcone gauge arbitrarily picks two of the coordinates, it makes Lorentz invariant not manifest. The Virasoro constraints, Eq. 6, in lightcone gauge become,

$$2p^+(\partial_\tau X^-) = (\partial_\tau X^i)^2 + (\partial_\sigma X^i)^2 \tag{9}$$

$$p^+(\partial_\sigma X^-) = (\partial_\tau X^i)(\partial_\sigma X^i) \tag{10}$$

with summation over $i = 1, \dots, D-2$ left implicit. So far we have fixed X^+ , and have found equations of X^- in terms of the other *transverse* coordinates. Therefore, the independent degrees of freedom are only those in $X^i(\sigma, \tau)$ which account for $2(D-2)$ functions to be found. At the classical level, it remains to solve for X^- .

Consider the open string for now, we may expand Eq. 7 as an infinite sum over modes with coefficients α_n^μ for each mode n as,

$$X^\mu = x^\mu + p^\mu \tau + i \sum_{n \neq 0} \frac{1}{n} \alpha_n^\mu e^{-in\tau} \cos n\sigma$$

where the $1/n$ factor was chosen for convenience. Using the definition of $X^- = \frac{1}{\sqrt{2}}(X^0 - X^{D-1})$ we write,

$$X^- = x^- + p^- \tau + i \sum_{n \neq 0} \frac{1}{n} \alpha_n^- e^{-in\tau} \cos n\sigma$$

We can solve for the coefficients α_n^- by plugging the last equation in the Virasoro constraints. Before doing that, it's easier to see that by adding and subtracting the two Virasoro constraints, we can bring them in the simpler form,

$$(\dot{X}^\mu \pm X'^\mu)^2 = 0 \tag{11}$$

where $\dot{X} = \partial_\tau X^\mu$ and $X' = \partial_\sigma X^\mu$. Expanding this in lightcone coordinates,

$$(\dot{X}^\mu \pm X'^\mu)^2 = (\dot{X}^i \pm X'^i)^2 - 2(\dot{X}^- \pm X'^-)(\dot{X}^+ \pm X'^+) = (\dot{X}^i \pm X'^i)^2 - 2p^+(\dot{X}^- \pm X'^-) = 0$$

Therefore,

$$(\dot{X}^- \pm X'^-) = \frac{1}{2p^+} (\dot{X}^i \pm X'^i)^2 = \frac{1}{2p^+} \sum_n \alpha_n^i e^{-in(\tau \pm \sigma)} \sum_m \alpha_m^i e^{-im(\tau \pm \sigma)}$$

with summation over $i = 1, \dots, D-2$ implicit again, and we have included the zero mode $\alpha_0^i = p^i$ and $\alpha_0^{+/-} = p^{+/-}$. Plugging in the same expansion for X^- the last equation reduces to,

$$\sum_k \alpha_k^i e^{-ik(\tau \pm \sigma)} = \sum_{n,m} \alpha_n^i \alpha_m^i e^{-i(n+m)(\tau \pm \sigma)}$$

and recalling that the exponential weights of the modes form a complete set of functions, we obtain,

$$\alpha_n^- = \frac{1}{2p^+} \sum_{m=-\infty}^{\infty} \alpha_{n-m}^i \alpha_m^i \quad (12)$$

For the closed string, the most general solution is a sum of two infinite series of independent modes, α and $\tilde{\alpha}$, corresponding to the left and right moving excitation. Very similar computations give us the two sets of modes in terms the transverse oscillations,

$$\alpha_n^- = \frac{1}{p^+} \sum_{m=-\infty}^{\infty} \alpha_{n-m}^i \alpha_m^i \quad (13)$$

$$\tilde{\alpha}_n^- = \frac{1}{p^+} \sum_{m=-\infty}^{\infty} \tilde{\alpha}_{n-m}^i \tilde{\alpha}_m^i \quad (14)$$

3 Quantisation in lightcone gauge

The lightcone gauge has allowed us to find the classical solution X^μ of the Polyakov action for the string that incorporates all of the Virasoro constraints. The next step is to quantise the theory by promoting the different modes to operators and imposing canonical commutation relations among them. We will then be able to extract the mass spectrum of the theory.

We can write the Polyakov action in the lightcone gauge explicitly,

$$\begin{aligned} S_P &= -\frac{1}{4\pi\alpha'} \int d^2\sigma \{(\partial_\tau X^\mu)(\partial_\tau X_\mu) - (\partial_\sigma X^\mu)(\partial_\sigma X_\mu)\} \\ &= \frac{1}{4\pi\alpha'} \int d^2\sigma \left\{ (\dot{X}^i)^2 - (X'^i)^2 \right\} - \frac{2}{4\pi\alpha'} \int d^2\sigma p^+ (\partial_\tau X^-) \end{aligned}$$

In this form we can directly identify the momentum conjugate to the X^i 's and $X^{+/-}$,

$$P_a^i = \frac{\partial \mathcal{L}_P}{\partial(\partial_a X^i)} = \frac{1}{2\pi\alpha'} \partial_a X^i, \quad \text{for } a = \tau, \sigma \text{ and } i = 1, \dots, D-2$$

Then, the canonical (and equal time, τ) commutation relations that we impose are the following,

$$\begin{aligned} [X^i(\sigma, \tau), P^j(\sigma', \tau)] &= i\delta^{ij}\delta(\sigma - \sigma') \\ [X^i(\sigma, \tau), X^j(\sigma', \tau)] &= [P^i(\sigma, \tau), P^j(\sigma', \tau)] = 0 \end{aligned}$$

Notice that as we claimed above, the physical degrees of freedom are only those in the X^i coordinates. Using the mode expansions that we have found above, we can compute (in the same way as in the second problem sheet) the commutation relations for the mode operators,

$$\begin{aligned} [x^i, p^j] &= i\delta^{ij}, & [x^+, p^-] &= [x^-, p^+] = -i \\ [\alpha_n^i, \alpha_m^j] &= [\tilde{\alpha}_n^i, \tilde{\alpha}_m^j] = in\delta_{m+n,0} \delta^{ij} \end{aligned} \quad (15)$$

and for the field operator X^μ to be hermitean, we get the constraint $(\alpha_n^\mu)^\dagger = \alpha_{-n}^\mu$.

Promoting the modes to quantum operators means that we need to take care of normal ordering issues in the sums over all the modes. This means that we want to order operators such that all annihilation operators are to the right of all creation operators. Notice however that this ambiguity arises only in sums involving terms of the form $\alpha_{-m} \cdot \alpha_m$. According to the commutation relations, we expect the normal ordering to be fixed by introducing an unknown constant a . Therefore, Eq. 12 becomes,

$$\alpha_n^- = \frac{1}{p^+} \left(\frac{1}{2} \sum_{i=1}^{D-2} \sum_{m=-\infty}^{\infty} : \alpha_{n-m}^i \alpha_m^i : - a \delta_{n,0} \right) \quad (16)$$

Consider the $n = 0$ case and recall that we have set $\alpha_0^i = p^i$ and $\alpha_0^- = p^-$,

$$\begin{aligned} p^- &= \frac{1}{p^+} \left\{ \frac{1}{2} \sum_{m \neq 0} : \alpha_{-m}^i \alpha_m^i : + \frac{1}{2} \alpha_0^i \alpha_0^i - a \right\} \\ &= \frac{1}{p^+} \left\{ \sum_{m=1}^{\infty} \alpha_{-m}^i \alpha_m^i + \frac{1}{2} (p^i)^2 - a \right\} \end{aligned}$$

On the other hand, when the string is on-shell, its mass is given by, $M^2 = -p^\mu p_\mu = 2p^+ p^- - (p^i)^2$ and we can write the above equation as what we call the *mass shell condition* of the open string,

$$M^2 = 2(N - a) \quad (17)$$

where we defined the number operator as $N = \sum_{n=1}^{\infty} \alpha_{-n}^i \alpha_n^i$. The massless condition shows that the mass of the string is completely determined by the transverse oscillations in the lightcone gauge. This should be contrasted with the covariant quantisation in which all the D different modes contributed to the mass and excitations. The corresponding mass shell condition for the closed string is

$$M^2 = 2(N + \tilde{N} - 2a) \quad (18)$$

and it is accompanied by a *level-matching* condition, $N = \tilde{N}$. A more elaborated treatment of this derivation is given in [1].

For later use we note that the mode operators satisfy a Virasoro algebra given by,

$$[p^+ \alpha_m^-, p^+ \alpha_n^-] = (m - n) p^+ \alpha_{-m}^- + n + \left\{ \frac{D-2}{12} (m^3 - m) + 2am \right\} \delta_{m+n,0}$$

Instead of carrying the lengthy calculation for the above relation, we just note that it follows by explicit computations using the commutation relations and the operator relation: $[AB, CD] = A[B, C]D + C[A, D]B + [A, C]BD + CA[B, D]$. The calculation is exactly along the lines of the calculation of the Virasoro generators commutation relation in the old covariant approach.

So far we succeeded in quantising the theory and we can obtain the spectrum by acting on the vacuum state with the creation operators. However, we started with a not manifestly covariant gauge and for consistency, we require that Lorentz invariance is respected. It turns that this is the case for a specific value of a and spacetime dimension D .

There is a direct way to calculate these by making use of the regularisation of the zeta function. One example of such a scheme is given by GSW[2], even though it is not the only one (see Chapter 3 by lecture notes of Prof. Weigand [1]). The argument proceeds by explicitly working out the normal ordering for modes i by using the commutation, $[\alpha_m^i, \alpha_{-m}^i] = m$,

$$\sum_{m=-\infty}^{\infty} \alpha_{-m}^i \alpha_m^i = \sum_{m=-\infty}^{\infty} : \alpha_{-m}^i \alpha_m^i : + \sum_{m=1}^{\infty} m$$

The last term is seemingly divergent and therefore following a common procedure used in QFT, we regularise it. This means that we can instead consider the sum $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$, which is the Riemann zeta function. Even though the function is only convergent for $Re(s) > 1$, we use that fact that it admits an analytic continuation to values with $Re(s) < 1$, and use directly the result that $\zeta(1) = -1/12$. Using this in Eq. 12, we get a $(D-2)$ factor coming from the sum over the transverse modes, and,

$$a = \frac{1}{2} \cdot \frac{D-2}{12} = \frac{D-2}{24}$$

We now have a relation of a in terms of the dimension D which we can use in our mass shell condition. As we will shortly see, studying the spectrum and requiring Lorentz invariance will force $a = 1$ and therefore $D = 26$.

4 Ensuring Lorentz Invariance

We will now follow a more careful treatment of the Lorentz invariance of the string in the lightcone quantisation by studying the generators of the Lorentz algebra. From the Noether procedure for obtaining the conserved current of a symmetry $X^\mu \rightarrow X^\mu + \delta X^\mu$ of a Lagrangian $\mathcal{L}$ with $\delta X^\mu = \epsilon^i F_i^\mu(X^\nu)$ and F_i^μ functions of the field X^μ , the conserved current j_i^μ is,

$$\epsilon^i j_i^\mu = \frac{\partial \mathcal{L}}{\partial_\mu (\partial_a X^l)} \delta X^l$$

The Polyakov action admits the translational symmetry, $X^\mu \rightarrow X^\mu + a^\mu$, and Lorentz symmetry, $X_\mu \rightarrow \epsilon \chi_{\mu\nu} X^\nu$ with $\chi_{\mu\nu} = -\chi_{\nu\mu}$. The currents associated to these are then correspondingly,

$$P_a^\mu = \partial_a X^\mu$$

$$J_a^{\mu\nu} = X^\mu \partial_a X^\nu - X^\nu \partial_a X^\mu$$

Given the currents, we can obtain the conserved charges by integrating over the range of the worldsheet coordinate σ . Substituting the mode expansions, we find the total momentum P^μ and total angular momentum J^μ are given by,

$$P^\mu = p^\mu, \quad J^{\mu\nu} = l^{\mu\nu} + E^{\mu\nu}$$

with $l^{\mu\nu} = x^\mu p^\nu - p^\mu x^\nu$ and $E^{\mu\nu} = -i \sum_{n=1}^{\infty} \frac{1}{n} (\alpha_{-n}^\mu \alpha_n^\nu - \alpha_{-n}^\nu \alpha_n^\mu)$.

We now focus on the Lorentz algebra generators $J^{\mu\nu}$, in the lightcone gauge, and specifically,

$$J^{i-} = x^i p^- - x^- p^i - i \sum_{n=1}^{\infty} \frac{1}{n} (\alpha_{-n}^i \alpha_n^- - \alpha_{-n}^- \alpha_n^i)$$

The reason for focusing on this is because the transverse J^{ij} charges will trivially be generators of the Lorentz algebra since they are not affected by the gauge choice. Even though this is not quadratic in the modes anymore, it is still true that the Poisson structure on the worldsheet is satisfied, namely, $\{J^{i-}, J^{j-}\} = 0$. Promoting the generators to quantum operators in normal ordered form, we require that the corresponding commutator is $[J^{i-}, J^{j-}] = 0$.

The commutator can be worked out directly by splitting it into pieces and use the commutation relation from above. This is rather lengthy and the full derivation will not add anything lightful to the report. It can be found in the Appendix B of the original paper of Goddard, Goldstone, Rebbi and Thorn[3] and is sketched in GSW[2]. The final result is,

$$[J^{i-}, J^{j-}] = \frac{1}{(p^+)^2} \sum_{m=1}^{\infty} \left[\left(\frac{D-26}{12} \right) m + \left(\frac{D-26}{12} + 2a - 2 \right) \frac{1}{m} \right] (\alpha_{-m}^i \alpha_m^j - \alpha_{-m}^j \alpha_m^i) \quad (19)$$

In the absence of a full derivation of the above commutation, an explanation is necessary. As can be seen from Eq. 12, α^- is a complicated sum over mode bilinears. The appearance of bilinears and, in fact, higher order products of modes in the classical version of the current under consideration means that the quantum version might come with *anomalies* that appear as extra terms in the quantum commutation relations. It turns out that the anomaly can only appear in terms quadratic in the operators α^i .

Requiring Lorentz invariance is equivalent to the vanishing of the commutation of the generators in Eq. 19. For the sum on the right hand side to vanish for all integers m , both terms in the bracket should vanish identically. Therefore the necessary and sufficient conditions for a Lorentz invariant theory in lightcone gauge appear as a critical dimension of the string, $D = 26$, and normal ordering constant $a = 1$. This is in agreement with the naive (at first sight) regularisation scheme implemented above.

Finally, let us note that this only holds for string propagation in flat Minkowski spacetime, and in fact for more complicated backgrounds, the Virasoro algebra might have a central charge that is different from the dimension of spacetime. This is nicely explained by Prof. Figueroa in a MathOverflow answer [4] in a more abstract algebraic context.

5 Mass Spectrum

We are now ready to study the mass spectrum of the bosonic string in lightcone quantisation. The Fock space is generated by defining the ground state $|0; k\rangle$ to be an eigenstate of the momentum operator with momentum k^μ and

$$\alpha_n^i |0; k^\mu\rangle = 0 \quad \text{for all } n > 0 \quad (20)$$

We can organise the states in terms of the eigenvalues of the number operator N , referred to as the level of the string. Starting with the open string, the mass shell condition with a replaced by the value that guarantees Lorentz invariance, becomes,

$$M^2 = 2(N - 1) = 2 \left(\sum_{n=0}^{\infty} \alpha_{-n}^i \alpha_n^i - 1 \right)$$

The mass shell condition for the ground state ($N = 0$) gives a negative norm, $M^2 = -2$. This means that the ground state is a tachyon and unfortunately, we cannot get around this in bosonic string theory. The same result was obtained with the covariant gauge, therefore we get at least a self-consistent theory. However, we note that this might mean that, similarly to what happens with spontaneously broken symmetries, the vacuum was chosen to be at an unstable (local) maximum which leads to a tachyonic mode.

To discuss excited states which are obtained by acting on the ground state with the creation operators of the transverse directions we consider again the Lorentz invariance. Before making the lightcone gauge, the theory has full $SO(1, D-1)$ Lorentz symmetry, and the lightcone gauge has an $SO(D-2)$ manifest Lorentz symmetry. However the excitations, being physical states, should still be irreducible representations of the $SO(1, D-1)$ group. To understand how this is the case, we recall the representation theory of the Poincare group. Massive particles are irreducible representations of the $SO(D-1)$ subgroup since there is a frame in which they can be described by a D component vector $(m, 0, \dots, 0)$.

With $N = 1$, the mass shell condition gives $M^2 = 0$, and therefore it is a massless state. In that case the above representation does not fit and we instead use a massless representation. The simplest description of a massless representation would be a $D-2$ component vector of the form $(p_1, p_1, 0, \dots, 0)$ and therefore the symmetry group is indeed $SO(D-2)$, as expected from the lightcone quantisation. There is only one such state and it lives in the **24** dimensional representation of $SO(24)$,

$$\alpha_{-1}^i |0, k^\mu\rangle$$

We now consider excitations at level two, namely $N = 2$, for the open string. The states come from two distinct representations,

$$\alpha_{-2}^i |0, k^\mu\rangle, \quad \alpha_{-1}^i \alpha_{-1}^j |0, k^\mu\rangle$$

From the mass shell condition we get that these are massive states with $M^2 = 2$. To find the representation of $SO(25)$ that the first excited states generate we can count its dimension; there are 24 states of the first kind and $\frac{24 \cdot 25}{2}$ of the second kind, since the state is symmetric in i, j . Therefore there is a total of 324 states and the representation is indeed the **324** dimensional one, which is precisely the symmetric and traceless second rank tensor of $SO(25)$. The second level excitation is said to be a massive spin-two particle.

Similarly, we discuss the spectrum of the closed string. The closed string mass shell condition is given by,

$$M^2 = 2(N + \tilde{N} - 2)$$

and at the same time, the level matching condition, which requires that there must be an equal number of right, α , and left, $\tilde{\alpha}$, moving excited modes, must be satisfied, $N = \tilde{N}$.

The ground state, $|0, 0, k^\mu\rangle$, is again a tachyon with mass $M^2 = -4$, and is in the trivial representation of $SO(1, 25)$. At level one, the two constraints allow for only states that are tensor products of the massless vector representations of $SO(24)$,

$$\alpha_{-1}^i \tilde{\alpha}_{-1}^j |0, 0, k^\mu\rangle$$

which gives a total of $(D-2)^2$ states and importantly it is massless, $M^2 = 0$. To study its representation, we note that we can decompose an $SO(24)$ tensor into,

$$T_{ij} = S_{(ij)} + A_{[ij]} + \delta_{ij} M$$

with S its symmetric and traceless part with dimension **299**, A its antisymmetric part with dimension **276** and M the trace of the tensor with dimension **1**. We say that the symmetric traceless part is a *graviton*, the antisymmetric part is a *dilaton* and the trace part is a scalar particle. The appearance of the graviton as a massless symmetric traceless tensor fits nicely with the expectation (proved by Weinberg) that a quantum theory of gravity should couple to conserved stress-energy tensor in the same way as in Einstein gravity[1].

Having established the spectrum up to level one, by self consistency, we can obtain all higher level states by taking tensor products of states in $SO(24)$ and $SO(25)$, depending the mass of the state, using Young Tableaux techniques. For example let us consider the spectrum of the closed string at level two. Denoting the ground state by $|0\rangle$ for brevity, there are now several possibilities,

$$\alpha_{-2}^i \tilde{\alpha}_{-2}^j |0\rangle, \quad \alpha_{-1}^i \alpha_{-1}^j \tilde{\alpha}_{-2}^k \tilde{\alpha}_{-2}^l |0\rangle, \quad \alpha_{-1}^i \alpha_{-1}^j \tilde{\alpha}_{-2}^k |0\rangle, \quad \alpha_{-1}^i \tilde{\alpha}_{-1}^j \tilde{\alpha}_{-2}^k |0\rangle$$

These states are massive with $M^2 = 4$. To obtain the representation content of these states we take the tensor of the symmetric and traceless representation of $SO(25)$, which is the **324** dimensional.

6 Constructing physical states

To study scattering amplitudes in string theory, we need to introduce *vertex operators* on the worldsheet, that correspond to absorption and emission of physical states. When working in the old covariant quantisation in the lectures, we discovered a correspondance between physical states and vertex operators. Our goal in this section is to construct these physical states, following the approach developed by Del Giudice, Di Vecchia and Fubini[3] (referred to as DDF).

Before discussing the DDF construction we need to define the Virasoro operators that were extensively used throughout the course in the OCQ formalism. The classical expression that define the Virasoro operators are (in normal ordered form),

$$L_m = \frac{1}{2} \sum_{n=-\infty}^{\infty} : \alpha_{m-n} \cdot \alpha_n : \\ \tilde{L}_m = \frac{1}{2} \sum_{n=-\infty}^{\infty} : \tilde{\alpha}_{m-n} \cdot \tilde{\alpha}_n :$$

At the classical level, the Virasoro operators satisfy the Witt algebra $\{L_m, L_n\}_{\text{P.B.}} = -i(m-n)L_{m+n}$. According to the prescription described for promoting classical modes to operators, we need to deal with the normal ordering issues. As explained above, these only arise in the expressions of L_0 and as before we fix this by introducing the normal ordering constant $L_0 \rightarrow L_0 - a$. The normal ordering issue alters the quantum commutation relation, and we find that the quantum Virasoro algebra becomes,

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0} \quad (21)$$

where c is the central charge that turns out to be D in our present case. The proof is rather lengthy and was carried out in the second problem set, so it omitted from this report.

We now proceed by defining a physical state of an open string $|\phi\rangle$ to be one for which $L_n |\phi\rangle = 0$ for all $n \in \mathbf{N}$ and $(L_n - a) |\phi\rangle = (L_0 - 1) |\phi\rangle$. The first constraint ensures that the modes with positive frequencies annihilate a physical state. The equivalent definitions for a closed string are $(L_0 - 1) |\phi\rangle = (\tilde{L}_0 - 1) |\phi\rangle = 0$.

We first recall that a local operator $A(\tau)$ defined on the worldsheet is said to have conformal dimension J if and only if, under the change of variables $\tau \rightarrow \tau'$, $A(\tau)$ transforms as,

$$A'(\tau') = \left(\frac{d\tau}{d\tau'} \right)^J A(\tau).$$

At the infinitesimal level, it turns out that since the operators L_m generate infinitesimal conformal transformations, we can show that in fact,

$$[L_m, A(\tau)] = e^{im\tau} \left(-i \frac{d}{d\tau} + mJ \right) A(\tau) \quad (22)$$

Assuming that the operator admits a Fourier expansion, $A(\tau) = \sum_{m=-\infty}^{\infty} A_m e^{-im\tau}$, by substituting above, making a relabeling in the sum and use the completeness of the functions $e^{-im\tau}$ we arrive to,

$$[L_m, A_n] = [m(J-1) - n] A_{m+n}$$

We also define a vertex operator for the emission of a state with momentum k^μ is defined by the normal ordered exponential,

$$V_k(\tau) =: e^{ik \cdot X(0, \tau)} :$$

Expanding the exponential we find terms of the form $k \cdot k$ which vanish for massless states and the normal ordering is in that case unnecessary. We then use a different form for the vertex operator of a massless state with a polarisation vector ζ given by

$$V_\zeta(k, \tau) = \zeta \cdot \frac{dX}{d\tau} e^{ik \cdot X}$$

We are now ready to build physical states as defined above. To do it, we will construct a set of transverse operators that commute with the Virasoro operators L_m , and have the property that when applied on the ground state, they generate physical states. First of all, it will be useful to restrict our discussion to *allowed states* which are defined by making an arbitrary but convenient choice as follows.

For the tachyon ground state, $|0, p_0\rangle$ with $p_0^\mu = (1, -1, 0, \dots, 0)$ (in the lightcone frame) and a null vector $k_0^\mu = (0, -1, \dots, 0)$ (i.e. $k_0^2 = 0$). Now, an allowed state with momentum p^μ is one for which when the mass shell condition is satisfied, $M^2 = 2(N-1)$, then $p^\mu = p_0^\mu - N k_0^\mu$.

Consider now a vector $k^\mu = n k_0^\mu$ for an integer n and its corresponding vertex operator, $V_\zeta(k, \tau)$. Singling out the transverse direction by choosing the i -th polarisation direction, we get,

$$V^i(nk_0, \tau) = \frac{dX^i}{d\tau} e^{inX^+(\tau)}$$

and now the fact that we have chosen k^μ to be an integer multiple of k_0^μ ensures that, when acting on allowed states, the whole operator is periodic with period 2π . This is why the notion of an allowed state is useful, and in fact not so restricting; by doing a Lorentz rotation of k_0^μ we can obtain all other states, except those with $p^\mu = 0$.

We define a DDF operator to be a Fourier mode in the expansion of V^i , which is guaranteed to exist by the periodicity of the operator. Thus,

$$A_n^i = \frac{1}{2\pi} \int_0^{2\pi} V^i(nk_0, \tau) d\tau \quad (23)$$

and by the hermiticity of X^μ , we automatically have $A_n^{i\dagger} = A_{-n}^i$ as is the case for the quantum modes, α_n . From Eq. 22, when the conformal dimension of an operator is $J = 1$, the commutation is a total derivative, namely,

$$[L_m, V(\tau)] = -i \frac{d}{d\tau} (e^{im\tau} V(\tau)).$$

We can compute the commutator between the L_m 's and the Fourier modes of a DDF operator to find that,

$$[L_m, A_n^i] = \frac{1}{2\pi} \int_0^{2\pi} [L_m, V^i(nk_0, \tau)] d\tau = -\frac{i}{2\pi} \int_0^{2\pi} \frac{d}{d\tau} [e^{im\tau} V^i(nk_0, \tau)] d\tau = 0$$

where the last equality follows because $e^{im\tau} V^i(nk_0, \tau)$ is 2π -periodic. This equation is helpful for building states that satisfy the Virasoro conditions since for $n = 0$ it reduces to $[N, A_n^i] = -n A_n^i$. DDF states are obtained by multiplying the tachyonic ground state $|0, p_0\rangle$ with the operators $A_{n_k}^{i_k}$,

$$|\phi, p_0 - Nk\rangle = A_{-n_1}^{i_1} A_{-n_2}^{i_2} \dots A_{-n_r}^{i_r} |0, p_0\rangle \quad (\dagger)$$

where ϕ denotes the partition of the level into the $D - 2$ transverse modes and the level is given by $N = \sum_k n_k$. From the vanishing of the commutator $[L_m, A_n^i]$ we conclude that DDF states satisfy the Virasoro conditions.

We now consider the time derivative of the transverse field operator $\dot{X}^i(\tau)$ at $\sigma = 0$ with mode expansion as before,

$$\dot{X}^i(\tau) = \sum_{n=-\infty}^{\infty} \alpha_n^i e^{-in\tau}$$

so that,

$$\begin{aligned} [\dot{X}^i(\tau_1), \dot{X}^j(\tau_2)] &= \sum_{n,m} e^{-in\tau_1} e^{-im\tau_2} [\alpha_n^i, \alpha_m^j] = i \sum_{n,m} e^{-in\tau_1} e^{-im\tau_2} m \delta^{ij} \delta_{m+n,0} \\ &= i \delta^{ij} \sum_m m e^{im(\tau_1 - \tau_2)} = 2\pi i \delta^{ij} \frac{d}{d\tau_1} \delta(\tau_1 - \tau_2) \end{aligned}$$

where on the last step we used a delta function representation. Therefore, we can compute the algebra between the DDF operators directly,

$$\begin{aligned} [A_m^i, A_n^j] &= \frac{1}{(2\pi)^2} \int_0^{2\pi} d\tau_1 \int_0^{2\pi} d\tau_2 [\dot{X}^i(\tau_1), \dot{X}^j(\tau_2)] e^{imX^+(\tau_1) + inX^+(\tau_2)} \\ &= \frac{i}{2\pi} \delta^{ij} \int_0^{2\pi} d\tau_1 \int_0^{2\pi} d\tau_2 \left(-\delta(\tau_1 - \tau_2) im \dot{X}^i(\tau_1) \right) e^{imX^+(\tau_1) + inX^+(\tau_2)} \\ &= \frac{m}{2\pi} \delta^{ij} \int_0^{2\pi} d\tau \dot{X}^+(\tau) e^{i(m+n)X^+(\tau)} \\ &= m \delta^{ij} \delta_{m+n,0} \end{aligned}$$

where on the second step we integrated by parts in τ_1 and on the last step we integrate exactly using $\dot{X}^+(\tau) = p^+ = 1$.

The above calculations show that the DDF operators satisfy the same algebra as the quantum mode operators and additionally they generate only physical states according to the definition given above, $L_n |\phi\rangle = 0$. Additionally, all the states of the form $\dagger$ have positive norm and can therefore be taken to be a representation of the physical Hilbert space. The only thing that is missing is to show that they in fact span the Hilbert space, a fact that is true when $D = 26$.

7 The no-ghost theorem

We have seen that if we build the Fock space using all the D -dimensional modes, as in the old covariant quantisation, we unavoidably end up with *ghosts* which are states of negative norm. Working in the lightcone gauge, we saw that the physical degrees of freedom are actually the transverse modes α_n^i for $i = 1, \dots, D-2$. Using this formalism, we built a set of DDF states with the property of non-negative norm, and therefore span the physical Hilbert space. We will now sketch the proof first given by Brower[5] and independently by Goddard and Thorn[6], that shows that DDF states indeed span the Hilbert space of physical states in $D = 26$ when $a = 1$ and the spectrum is thus ghost-free. We will closely follow the paper by Goddard and Thorn building on the DDF formalism developed above.

We define a *spurious state* $|\phi\rangle$ to be a state which satisfies $(L_0 - 1)|\phi\rangle = 0$ and $\langle\psi|\phi\rangle = 0$ for all the physical states $|\psi\rangle$. Spurious states can be written as linear combinations in the form,

$$|\phi\rangle = \sum_{n=1}^{\infty} L_{-n} |\chi_n\rangle$$

where $|\chi_n\rangle$ is such that $(L_0 - 1 + n)|\chi_n\rangle$. A way to understand this is by recalling first that a physical state is a highest weight representation of the Virasoro algebra since $L_n|\psi\rangle = 0$ for all $n > 0$. By definition, a spurious state $|\chi_n\rangle$ satisfies $\langle\chi_n|(L_n|\psi\rangle) = 0$, since $L_n|\psi\rangle$ is also physical. We can rewrite the last equation as

$$0 = \langle\chi_n|L_n|\psi\rangle = \langle\psi_n|L_n^\dagger|\chi\rangle = \langle\psi|L_{-n}|\chi_n\rangle = \langle\psi|(L_{-n}|\chi_n\rangle)$$

which shows that the state $L_{-n}|\chi_n\rangle$ is orthogonal to the physical state $|\psi\rangle$ and it is therefore spurious. A state that is both spurious and physical is orthogonal to all other physical states (and therefore to itself), so it has a vanishing norm and we call it *null*.

We are now ready to prove the no-ghost theorem that states that a solution to the Virasoro physicality conditions $L_n|\psi\rangle = 0$ and $(L_0 - 1)|\psi\rangle = 0$ in $D = 26$ dimensions is of the form $|\psi\rangle = |f\rangle + |ns\rangle$, where $|f\rangle$ is a DDF state and $|ns\rangle$ is a null state. To prove it we use the properties of the DDF states constructed in the section above. In addition, we define $K_n = k \cdot \alpha_n$, so that $K_n|f\rangle = 0$ with $K_n = K_{-n}^\dagger$ and it satisfies the algebra,

$$[K_m, K_n] = 0, \quad [L_m, K_n] = -nK_{m+n}$$

The first one follows from the choice of k being lightlike, $k^2 = 0$, and the second is a direct consequence of,

$$[\alpha_m^\mu, L_n] = \frac{1}{2} \sum_{r=-\infty}^{\infty} [\alpha_m^\mu, : \alpha_r \cdot \alpha_{n-r} :] = \frac{1}{2} \sum_{r=-\infty}^{\infty} (m\delta_{m+r,0}\alpha_{n-r}^\mu + m\delta_{m+n-r,0}\alpha_r^\mu) = m\alpha_{n+m}^\mu$$

Consider now an orthonormal basis for the DDF states denoted by $|f, M, \kappa\rangle$ where M is the mass of the string and κ counts the degeneracy of the string. The first step in the proof shows that a generic subspace of the whole Hilbert space of the theory that is given by the level $N = M + \sum_r \lambda_r r + \sum_s \mu_s s$ has a basis that can be written as,

$$L_{-1}^{\lambda_1} L_{-2}^{\lambda_2} \dots L_{-n}^{\lambda_n} K_{-1}^{\mu_1} K_{-2}^{\mu_2} \dots K_{-n}^{\mu_m} |f, M, \kappa\rangle \quad (\ddagger)$$

The proof that the above is indeed a basis of DDF states requires showing that they are independent and that they span the space. However, by dimensional counting of the set of the basis and the dimension of the subspace, independence is the only non-trivial bit. The proof is

given in the Appendix of [6]. It involves constructing the smaller subspaces of the big subspace so that the state is a linear combination of the various $|f, M, \kappa\rangle$ starting from the vacuum state and then look individually at the highest excitations of each level.

Before completing the proof of the no-ghost theorem, we need a secondary result that will help us write a spurious state in a convenient basis. Let $\mathcal{S}^{l_0}$ be the space of spurious states with $L_0 = l_0$ and $\tilde{L}_2 = L_2 + \frac{3}{2}L_1^2$. The claim to prove is that the following maps hold only when $D = 26$,

$$L_1 : \mathcal{S}^1 \rightarrow \mathcal{S}^0, \quad \tilde{L}_2 : \mathcal{S}^1 \rightarrow \mathcal{S}^{-1}$$

The proof is based on the fact that the Virasoro operators with $n \geq -1$ can be generated by only L_{-1} and L_{-2} . This is because of the commutation relation,

$$L_{-1}L_{-2} = [L_{-1}, L_{-2}] + L_{-2}L_{-1} = L_{-3} + L_{-2}L_{-1}$$

which we can successively use to generate all L_{-n} with $n \geq -1$. Consider the states $|\lambda, n\rangle$ such that $L_0 |\lambda, n\rangle = n |\lambda, n\rangle$. Instead of using L_{-2} we can perform a change of basis so that a generic spurious state $|s\rangle \in \mathcal{S}^1$ can be written as,

$$|s\rangle = L_{-1} |\lambda, 0\rangle + \tilde{L}_{-2} |\lambda, -1\rangle$$

Using the Virasoro algebra relations, we find that,

$$\begin{aligned} L_1 |s\rangle &= L_{-1}L_1 |\lambda, 0\rangle + \tilde{L}_{-2}L_1 |\lambda, -1\rangle \\ \tilde{L}_2 |s\rangle &= |s'\rangle + \frac{D-26}{2} |\lambda, -1\rangle \end{aligned}$$

where $|s'\rangle \in \mathcal{S}^{-1}$. The D dependence of the second relation comes from $[L_2, L_{-2}] = 4L_0 + \frac{D}{2}$. We immediately see that $L_1 |s\rangle \in \mathcal{S}^1$ but $\tilde{L}_2 |s\rangle \in \mathcal{S}^{-1}$ only if $D = 26$, and the claim is proved.

We further define the space $\mathcal{K}^{l_0}$ of states with basis $K_{-1}^{\mu_1} K_{-2}^{\mu_2} \dots K_{-n}^{\mu_m} |f, M, \kappa\rangle$ that satisfy the condition $L_0 = l_0$. Since any state of the form of $\ddagger$ is either spurious (when there is a $\lambda_i \neq 0$) or it lies in $\mathcal{K}^{l_0}$. From the previous result these states provide a basis for the Hilbert space of physical states, we can decompose any state $|\psi\rangle$ that is physical as,

$$|\psi\rangle = |\phi\rangle + |s\rangle$$

where $|s\rangle$ is spurious and $|\phi\rangle \in \mathcal{K}^{l_0}$. Linear independence of the states in $\ddagger$ ensures that if $|\psi\rangle \in \mathcal{S}^{l_0}$ then we must have $|s\rangle, |\phi\rangle \in \mathcal{S}^{l_0}$ individually.

We now show that both $|\phi\rangle$ and $|s\rangle$ are physical, $L_m |\phi\rangle = L_m |s\rangle = 0$ for all $m \geq 0$. This is where we will use the intermediate result above which ensures that the action of L_0 maps states in $\mathcal{S}^1$ into $\mathcal{S}^0$ and $\mathcal{K}^1$ into $\mathcal{K}^0$. From the independence of $|\phi\rangle$ and $|s\rangle$ we must have $L_1 |\phi\rangle = L_1 |s\rangle = 0$ and $\tilde{L}_2 |\phi\rangle = \tilde{L}_1 |s\rangle = 0$. By the same argument that $L_1, \tilde{L}_2$ generate all other operators, it must then be true that indeed $L_m |\phi\rangle = L_m |s\rangle = 0$ for all $m \geq 0$.

The above argument shows that $|s\rangle$ is physical and spurious and therefore null. This means that $\langle s|s\rangle = \langle s|\phi\rangle = 0$ and thus, $\langle \psi|\psi\rangle = \langle \phi|\phi\rangle$. It remains to prove that $|\phi\rangle$ is a DDF state, which we prove by showing that its norm is non-negative. Since $|\phi\rangle \in \mathcal{K}^{l_0}$, we can write,

$$\begin{aligned} |\phi\rangle &= \sum_{\{\mu\}} C_{\{\mu\}}^{\kappa M} K_{-1}^{\mu_1} K_{-2}^{\mu_2} \dots K_{-n}^{\mu_m} |f, M, \kappa\rangle \\ &= |f', M, \kappa\rangle + \sum_{\{\rho\}} C_{\{\rho\}}^{\kappa M} K_{-1}^{\rho_1} K_{-2}^{\rho_2} \dots K_{-n}^{\rho_m} |f, M, \kappa\rangle \\ &= |f', M, \kappa\rangle + |\chi\rangle \end{aligned}$$

where $C_{\{\mu\}}^{\kappa M}$ are the coefficients and $\{\mu\}$ is a subset of the indices that make up the state. In the second step we split the state into the trivially DDF part, $|f', M, \kappa\rangle$ (which could happen to be 0) and summed over the reduced set of indices $\{\rho\}$. Using $[K_m, K_n] = 0$ and $K_n = K_{-n}^\dagger$, and commuting all the K_n 's with the K_{-n} , it is easy to see that, $\langle\chi|\chi\rangle = 0$, from which we immediately conclude that $\langle\phi|\phi\rangle = \langle f', M, \kappa|f', M, \kappa\rangle \geq 0$, with $\langle f', M, \kappa|f', M, \kappa\rangle = 0$ if the state is the vacuum. This completes the proof that in $D = 26$, the Hilbert space of physical states contains no ghosts.

8 Concluding Remarks

We introduced the Nambu-Goto and Polyakov action to solve for the equation of motion of a two dimensional worldsheet embedded in a D -dimensional Minkowski spacetime and found that the latter is easier to solve and to quantise. We solved the for the classical equation of motion of the string and found that it is complemented by two non-linear constraints, the *Virasoro constraints* that need to be taken care in the quantised theory.

After exploring the symmetries of the action, we found that the lightcone gauge is very convenient since it allows to solve the EOM and the constraints at the classical level. This showed that the string dynamics depend on the transverse modes, which account for only $D - 2$ degrees of freedom. The drawback was that the gauge was not manifestly covariant and therefore we had to prove this independently.

In quantising the theory, we faced an ambiguity in the normal ordering of the creating and annihilation operators and used it to conclude that Lorentz invariance is only possible in $D = 26$ dimensions. We further analysed the spectrum and discovered that the graviton is in fact an excitation mode of the closed string at level two.

Finally, we discussed how to get rid of ghost states, which are unphysical states with negative norm, by building the DDF states. We then used the DDF states to prove the no-ghost theorem which asserts that the spectrum of the theory is free of ghosts if the dimension of spacetime is in fact $D = 26$.

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